Disaster Recovery Runbook

Manufacturing ERP System

Date: January 9, 2026

Client: Manufacturer

Estimated Duration: ~3 Weeks

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1) Executive Summary

This runbook outlines failover and failback procedures for the Manufacturing ERP System in Azure, including planned and unplanned failover scenarios. The document follows best practices for DR architecture, with Azure Site Recovery (ASR) and Front Door ensuring minimal downtime and high availability.

DR Architecture Summary

Component	Primary Region	DR Region
Location	Italy North	North Europe
Hub VNet	10.0.0.0/16	10.10.0.0/16
SQL Server VM	vm-sql-erp-italynorth	vm-sql-erp-italynorth (replica)
Web Server VM	vm-web-erp-italynorth	vm-web-erp-italynorth (replica)
Recovery Vault	-	rsv-erp-dr-northeurope

Recovery Objectives

Metric	Target	Actual
RPO (Recovery Point Objective)	15 minutes	~5-15 minutes
RTO (Recovery Time Objective)	60 minutes	~30-45 minutes
Front Door Failover	Automatic	~30 seconds health probe

2) Contact Information

Escalation Matrix

Level	Role	Contact	Availability
L1	Cloud Architect	Abdelrahman	24/7
L2	Cloud Architect	Ziad	Business Hours
L3	Cloud Architect	Mohamed	On-Call

Azure Support

Support Type	Contact Method
Azure Support Ticket	portal.azure.com → Help + Support
Azure Status	status.azure.com
Service Health	Azure Portal → Service Health

3) Pre-Requisites

3.1 Access Requirements

Requirement	Details
Azure Portal Access	Contributor role
VPN Client	Configured with certificate
Recovery Services	Contributor role on vault

4) Disaster Classification

Level	Description	Response Time	Failover?
SEV 1	Complete region outage	Immediate	Full
SEV 2	Multiple service failures	15 minutes	Partial
SEV 3	Single VM/service failure	30 minutes	Consider
SEV 4	Performance degradation	1 hour	No

5) Failover Procedures

5.1 Scenario A: Test Failover (DR Drill – No Production Impact)

1. Create Test Failover
    - Recovery Services vault → Replicated items → select VM → Test Failover
    - Recovery Point: Latest processed
    - Azure VNet: vnet-dr-test-isolated (10.100.0.0/16)
2. Verify Test VMs
3. Run DR Validation Tests

Test	Expected Result	Actual	Pass/Fail
SQL VM boots	Running state		
Web VM boots	Running state		
IIS service running	W3SVC started		
DB accessible	Can query database		

4. Cleanup Test Failover

5.2 Scenario B: Full Region Failover (Unplanned)

Trigger: Primary region (Italy North) is unavailable
Estimated Time: 30-45 minutes

Step 1: Verify Outage (5 minutes)

```
# Check Azure Service Health
az monitor activity-log list --resource-group "rg-erp-hub-italynorth" --max-
events 10 -o table

# Check VM status in primary region
az vm list -d --query
"[?location=='italynorth'].{Name:name,State:powerState}" -o table

# Check Front Door origin health
az afd origin show --profile-name "fd-erp-manufacturing" --resource-group
"rg-erp-shared-italynorth" --origin-group-name "og-erp-webapp" --origin-name
"origin-firewall-dnat" --query "{Name:name,Enabled:enabledState}" -o table
```

Step 2: Initiate ASR Failover (15-20 minutes)

Portal Method:

1. Go to Recovery Services vaults → rsv-erp-dr-northeurope
2. Select vm-sql-erp-italynorth and click Failover
3. Configure failover settings (select Recovery Point and Azure VNet for DR)
4. Repeat for vm-web-erp-italynorth (Web Server)

step 3: Verify Failover Completion (5-10 minutes)

```
# Check VM status in DR region
az vm list -d --query
"[?location=='northeurope'].{Name:name,State:powerState,PrivateIP:privateIps}
" -o table

# Verify VMs are running
az vm show -g "RG-ERP-SPOKE-DB-ITALYNORTH-ASR" -n "vm-sql-erp-italynorth" --
show-details --query "{Name:name,State:powerState,PrivateIP:privateIps}" -o
table
```

Step 4: Verify Network Connectivity (5 minutes)

```
# Verify DR Firewall is running
az network firewall show --resource-group "rg-erp-hub-dr-northeurope" --name
"fw-erp-dr-northeurope" --query
"{Name:name,ProvisioningState:provisioningState}" -o table

# Verify route tables are associated
az network vnet subnet show --resource-group "rg-erp-spoke-db-italynorth-asr"
--vnet-name "vnet-spoke-db-italynorth-asr" --name "snet-sql" --query
"{Name:name,RouteTable:routeTable.id}" -o table
```

Step 5: Verify Front Door Routing (Automatic)

```
# Check Front Door origins
az afd origin list --profile-name "fd-erp-manufacturing" --resource-group
"rg-erp-shared-italynorth" --origin-group-name "og-erp-webapp" --query
"[].{Name:name,Host:hostName,Priority:priority,Enabled:enabledState}" -o
table
```

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6) Failback Procedures

6.1 Prerequisites for Failback

- Primary region is healthy
- Primary VMs are stopped or deleted
- ASR replication is set up for reverse replication

Step 1: Verify Primary Region Health

```
# Check primary region services
az network vnet show --resource-group "rg-erp-hub-italynorth" --name "vnet-
hub-italynorth" --query "provisioningState" -o tsv

az network firewall show --resource-group "rg-erp-hub-italynorth" --name "fw-
erp-italynorth" --query "provisioningState" -o tsv
```

Step 2: Re-Protect VMs (Enable Reverse Replication)

1. Go to **Recovery Services vault → Replicated items**
2. Select VM and click **Re-protect**

Step 3: Initiate Planned Failback

1. Once replication is healthy, click **Planned Failover** (not regular failover)
2. Follow normal failover steps to return to the primary region

Step 4: Commit Failback

1. Verify application works from primary region
2. Commit failover to complete failback
3. Re-enable replication to DR region

7) Single Service Recovery

7.1 VM Restart Procedure

```
# Restart SQL VM
az vm restart --resource-group "rg-erp-spoke-db-italynorth" --name "vm-
sql-erp-italynorth"

# Restart Web VM
az vm restart --resource-group "rg-erp-spoke-app-italynorth" --name "vm-
web-erp-italynorth"
```

7.2 Azure Firewall Recovery

- Check firewall provisioning/health
- If deallocated/misconfigured, reallocate/restore configuration
- Validate DNAT/network rules and outbound connectivity

7.3 VPN Gateway Recovery

- Check VPN gateway provisioning/health
- If tunnel down, validate local network gateway settings + connection status
- Reset gateway only if required

8) Validation Checklist (After failover)

- VMs are running in DR and services are up (SQL service, IIS service).
- App connectivity: Web → SQL (1433) and app returns expected HTTP response.
- Confirm Front Door endpoint is serving traffic from DR